Academic Year: 2025-26

Sap Id:60003240292

Shri Vile Parle Kelavani Mandal's
DWARKADAS J. SANGHVI COLLEGE OF ENGINEERING
(Autonomous College Affiliated to the University of Mumbai)
NAAC Accredited with "A" Grade (CGPA : 3.18)

Department of Information Technology

COURSE CODE: DJS23ILPC501 **DATE:** 23-09-2026

COURSE NAME: Computer Networks **CLASS: T. Y. B. Tech**

Experiment No. 6

CO/LO: Analyse various concerns at different layers in reference models using suitable techniques.

Aim: 1. To perform VLSM in packet tracer

2. Complete Module 14 of **CCNA**

Theory:

Subnetting: The idea of splitting a block of any IP address class to smaller blocks is referred to as subnetting. In **subnetting,** a network is divided into several smaller subnetworks (subnets) with each subnetwork having its own subnetwork address.

There are two types of subnetting:

1) FLSM (Fixed-Length Subnet Masking)
2) VLSM (Variable-Length Subnet Masking)

Fixed-Length Subnet Masking (FLSM) creates subnets all of the same size. But where some subnets will have many hosts and some have few, FLSM results in some subnets where number of required hosts is very much less than the number of hosts in the subnet. This results into wastage of IP addresses. The concept of VLSM has been introduced to avoid the wastage of IP addresses in FLSM.

Variable-Length Subnet Masking (VLSM) allows network engineers to divide an IP address space into a hierarchy of subnets of different sizes, making it possible to create subnets with very different host counts without wasting large numbers of addresses.

Problem statement:

A Company has three departments connected with WAN links. The company has

- The development department which requires 74 computers.
- Production department which requires 52 computers.
- Administrative department which requires 28 computers.
- All departments are connected with each other via wan link.
- Each wan link requires two IP addresses.

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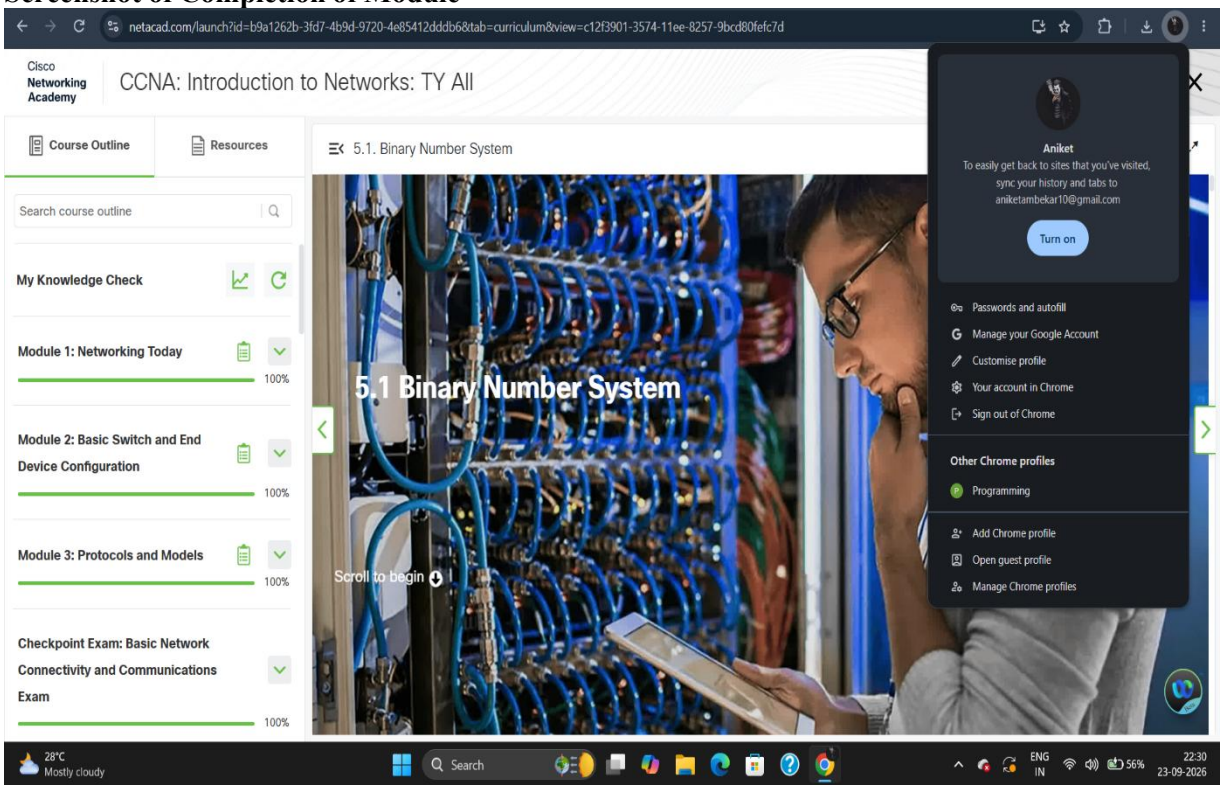
The company has been given an address space of 192.168.1.0/24. Design subnets according to the requirements of the company.

Steps for VLSM Subnetting

- Find the largest segment. Segment which needs largest number of host address.
- Do subnetting to fulfill the requirement of largest segment.
- Assign the appropriate subnet mask for the largest segment.
- For the second largest segment, take one of these newly created subnets and apply a different, more appropriate, subnet mask to it.
- Assign the appropriate subnet mask for the second largest segment.
- Repeat this process until the last network.
- Use static routing to connect WAN links.
- Use ping command to check connectivity in each subnetwork.

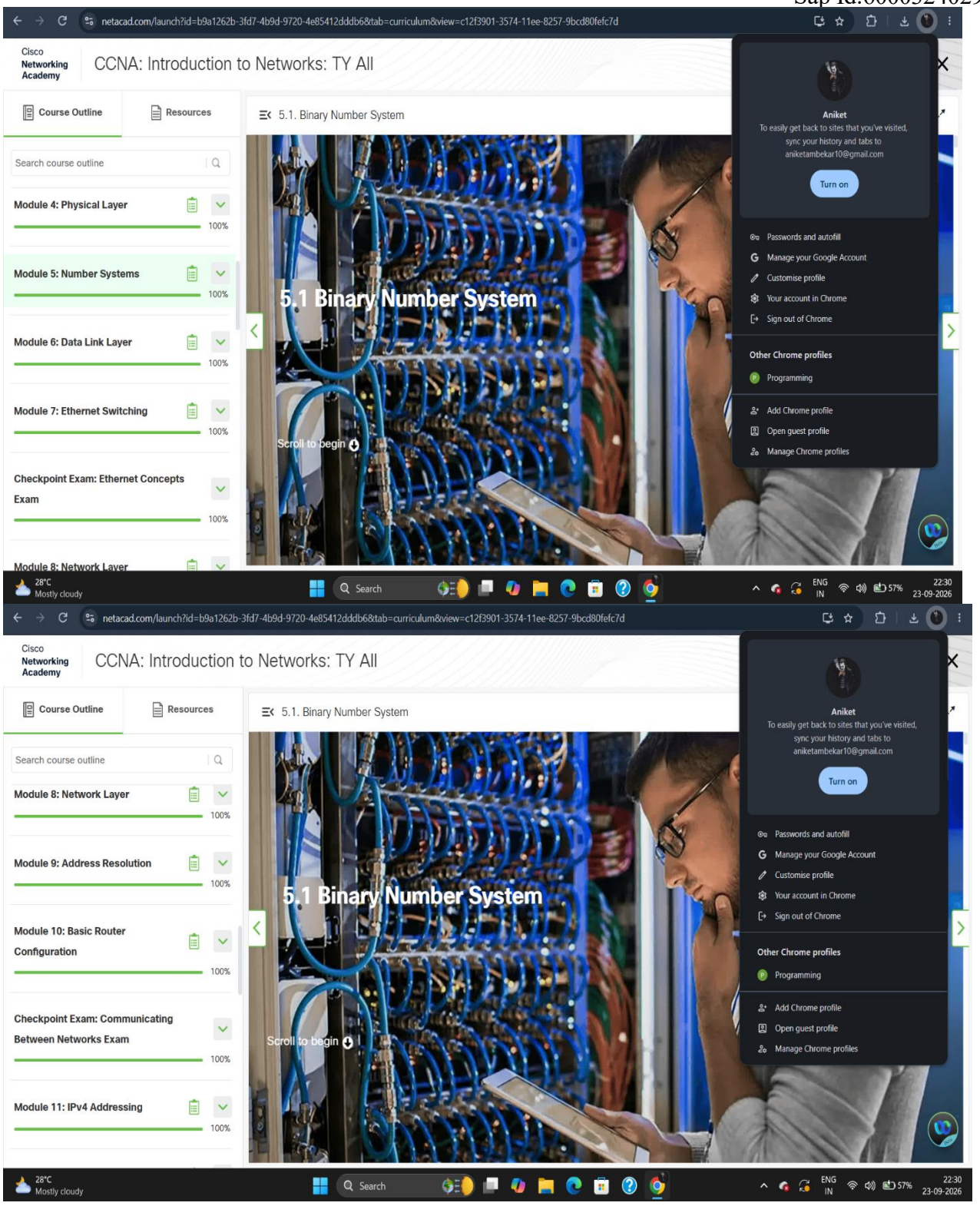
Procedure:

1. **Screenshot of Completion of Module**



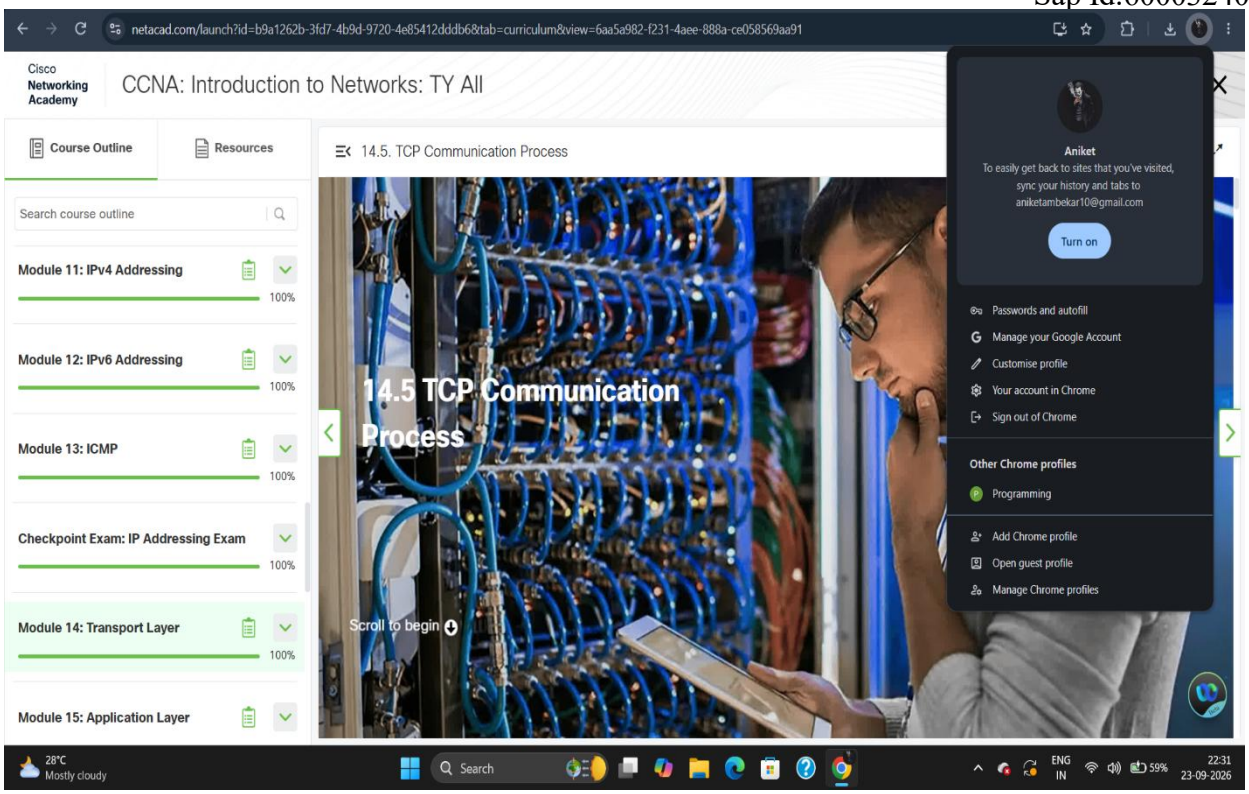
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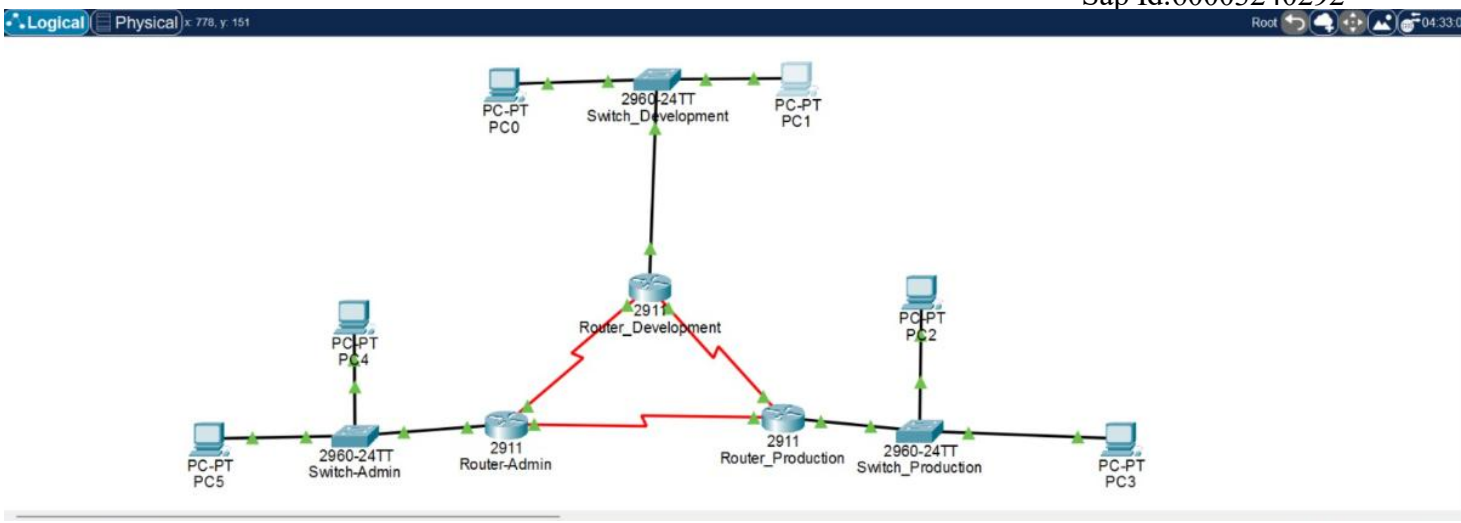


2. **Create six subnets three for departments (Three LAN's and three for WAN links)**

   Subnet Designing: Sheet to be attached

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C:\>exit

C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

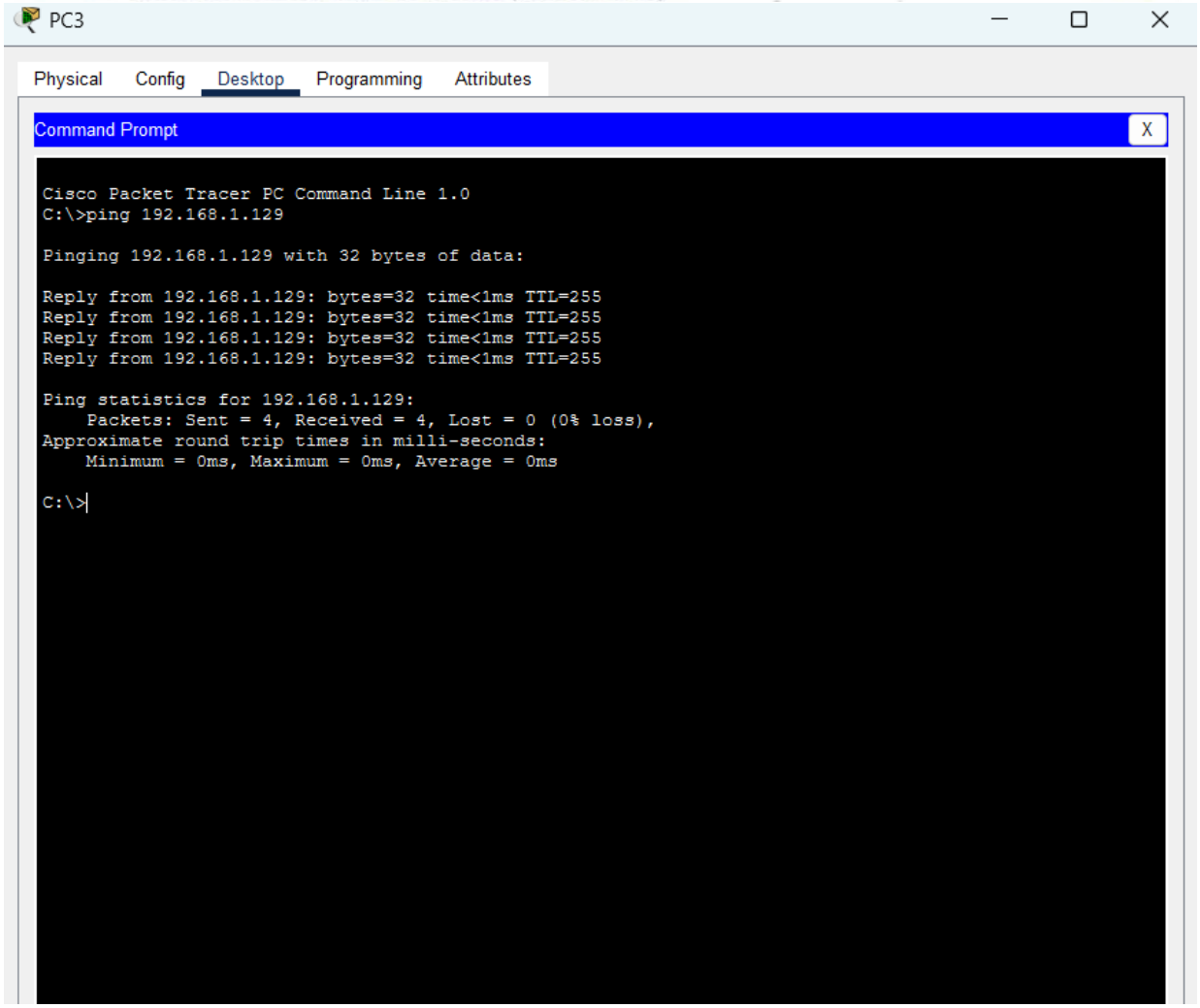
Reply from 192.168.1.1: bytes=32 time=5ms TTL=255
Reply from 192.168.1.1: bytes=32 time=9ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 9ms, Average = 3ms

C:\>
```

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Subnet	IP address require-ment	CIDR	Network address (net id)	First host IP address	Last host IP address	Broadcast IP address	Subnet mask
Subnet 1	74	/25	192.168.1.0	192.168.1.1	192.168.1.126	192.168.1.127	255.255.255.128
Subnet 2	52	/26	192.168.1.128	192.168.1.129	192.168.1.190	192.168.1.191	255.255.255.192
Subnet 3	28	/27	192.168.1.192	192.168.1.193	192.168.1.222	192.168.1.223	255.255.255.224
Subnet 4	2	/30	192.168.1.224	192.168.1.225	192.168.1.226	192.168.1.227	255.255.255.252





Subnet 5	2	/30	192.168.1.228	192.168.1.229	192.168.1.230	192.168.1.231	255.255.255.252
Subnet 6	2	/30	192.168.1.232	192.168.1.233	192.168.1.234	192.168.1.235	255.255.255.252

Conclusions:

Observations

1. **You are given the IP address block 192.168.10.0/24 to allocate to the following departments in a company:**
   i. HR Department: 50 hosts
   ii. Finance Department: 25 hosts
   iii. IT Department: 12 hosts
   iv. Admin Department: 6 hosts

2. Using Variable Length Subnet Masking (VLSM), perform the following tasks:
   i. Determine the subnet mask required for each department.
   ii. Assign appropriate subnets from the given block to each department.
   iii. Show the range of usable IP addresses for each subnet.
   iv. Identify any unused IP addresses